Profit-Driven Credit Approval Thresholding

From Predictions to Portfolio Outcomes

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Data Science Project Report

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A data science project exploring the intersection of machine learning and financial decision-making

Executive Summary

This project examines a practical question in consumer lending: **which loan applicants should a lender approve to maximize net profit?** Credit risk models are often evaluated on predictive accuracy, but in a lending context the business objective is profitability. This report presents a framework for optimizing the approval threshold of a probabilistic classifier with portfolio profit as the primary objective. Using publicly available LendingClub loan data from 2007 to 2018 (1.34 million loans), we trained Logistic Regression and Random Forest models across four feature engineering plans. Approval thresholds were systematically swept from 0.05 to 0.95 in increments of 0.01, and realized portfolio profit was calculated at each threshold using historical cash flows. **Key Observations:**

- **Best performing model:** Logistic Regression on 10 baseline features with AUC = 0.6671
- **Optimal threshold identified:** 0.620 — reject loans with default probability above 62%
- **Portfolio profit at this threshold:** $123,795,944 on the test set
- **Test Set Performance:** Reduced losses by $280k vs. Approve-All during the 2016-2018 adverse period

This framework suggests that the threshold—not the model alone—has a meaningful impact on profitability. A well-calibrated model with moderate discriminative power can be more useful for decision-making than a higher-AUC model with less reliable probability estimates. The results reinforce the value of aligning model evaluation with business objectives.

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1. Introduction

Credit scoring models are widely used in consumer lending to estimate the probability that a borrower will default on a loan. These models are often evaluated using statistical metrics such as AUC and accuracy, which measure how well the model separates good loans from bad ones. However, in a lending context, the relevant business question is profitability, not statistical performance. This project explores a practical question: **which loan applicants should a lender approve to maximize net profit?** A predicted probability of default does not by itself determine a decision. Some cutoff must be chosen—above which a loan is rejected, below which it is approved. The optimal cutoff depends on factors such as interest rates, loan amounts, recovery rates, and the distribution of default probabilities in the portfolio. **Project Objectives:**

- Train probabilistic classifiers to estimate default likelihood for each loan
- Systematically sweep approval thresholds from 0.05 to 0.95
- Calculate realized portfolio profit at each threshold using historical outcomes
- Identify the threshold that maximizes portfolio profit
- Compare against simple heuristic baselines to quantify value added

A central observation from this project is that the threshold choice has a material impact on profitability. A well-calibrated model with moderate discriminative power can support better financial decisions than a higher-AUC model with less reliable probability estimates. This underscores the importance of calibration and business alignment in applied data science work.

2. Methodology

2.1 Data

We used publicly available LendingClub loan data spanning 2007–2018, accessible through Kaggle. The dataset contains **1.34 million** loan records with 110 features covering borrower characteristics, loan terms, and eventual outcomes. **Data Cleaning:**

- Removed loans with incomplete or inconsistent status information
- Converted categorical variables (term, grade, home ownership) to numeric representations
- Dropped sparse columns (missing rate > 80%)
- Removed 25 columns containing payment history, collections, or post-origination behavior

Target Definition: The binary target *default* was defined as loans with status 'Charged Off' (1) versus 'Fully Paid' or 'Current' (0).

Cash Surplus Calculation: For each loan, realized net cash flow was computed as:

```
Cash Surplus = total_received - funded_amount
```

Where *total_received* includes all payments made by the borrower, and *funded_amount* is the principal disbursed. The total portfolio net cash surplus across all loans was **$555.6 million**.

Train/Test Split: Data was split 80/20 by issue date (temporal split) to prevent look-ahead bias and simulate realistic deployment conditions.

2.2 Feature Engineering

We developed four progressively complex feature engineering plans, all implemented through a custom *SafeFeatureEngineer* class that enforces strict train/test separation. All aggregations were calculated exclusively on the training set and then applied to the test set to prevent data leakage.

Plan	Features	Description
Baseline Minimal	10	Interpretable features directly from application data
Domain Enhanced	17	Adds credit risk domain knowledge features
Interaction Heavy	24	Adds interaction terms and log transformations
ML-Informed	37	Adds default rates by category and polynomial features

2.3 Model Training and Calibration

We trained two widely-used probabilistic classifiers on each of the four feature engineering plans: **Logistic Regression:** A linear model with L2 regularization (max iterations = 1000). We applied Platt scaling (sigmoid calibration) with 5-fold cross-validation to ensure probability estimates accurately reflect true default likelihood. **Random Forest:** An ensemble of 100 decision trees (max depth = 10). We applied isotonic regression calibration with 5-fold cross-validation to transform raw prediction scores into well-calibrated probabilities. Both models were trained with a fixed random seed (42) for reproducibility.

2.4 Threshold Optimization

For each trained model, we performed a systematic threshold sweep from 0.05 to 0.95 in increments of 0.01. At each threshold value, we applied the decision rule:

```
Approve loan if predicted_default_probability < threshold
```

The total portfolio profit at each threshold was computed by summing the realized profits of all approved loans. The optimal threshold was selected as the value that maximized this total portfolio profit.

2.5 Baseline Strategies

To contextualize our model performance, we compared against 8 baseline strategies:

- Approve-All: Approve every loan (no filtering)
- Reject-All: Approve no loans (profit = $0)
- Random (50%): Randomly approve half the loans
- Fixed Threshold 0.30: Approve loans with default probability < 0.30
- Fixed Threshold 0.50: Approve loans with default probability < 0.50
- Fixed Threshold 0.70: Approve loans with default probability < 0.70
- Grade-Based (D-F): Approve only loans with grade A-C
- DTI-Based (≤30%): Approve only loans with DTI ≤ 30%
- FICO-Based (≥660): Approve only loans with FICO ≥ 660

2.6 Target Leakage Detection

A critical finding during this project was the presence of target leakage in the original feature set. Initial model training produced an AUC of **0.9999**—a suspiciously high value indicating that features containing information about post-origination behavior were included in the training data. We identified 25 columns containing patterns matching payment history, collections, or post-origination behavior. These columns were systematically removed from both train and test sets. After removal, the AUC dropped from **0.9999** to a realistic **0.6671**, confirming that leakage had been effectively eliminated.

3. Results

3.1 Model Performance Comparison

Feature Plan	Model	AUC	Brier	Optimal Threshold	Optimal Profit
Baseline Minimal	Logistic Regression	0.6671	0.1622	0.620	**$123,795,944**
Baseline Minimal	Random Forest	0.7001	0.1558	0.920	$123,790,408
Domain Enhanced	Logistic Regression	0.6524	0.1639	0.890	$123,790,408
Domain Enhanced	Random Forest	0.6997	0.1558	0.870	$123,790,408
Interaction Heavy	Logistic Regression	0.6805	0.1610	0.940	$123,679,150
Interaction Heavy	Random Forest	0.7002	0.1558	0.910	$123,790,408
ML Informed	Logistic Regression	0.6794	0.1610	0.930	$123,677,627
ML Informed	Random Forest	0.6999	0.1570	0.790	$123,790,408

Table 1: Model performance metrics across all feature plans and model types. Best performing model highlighted in blue.

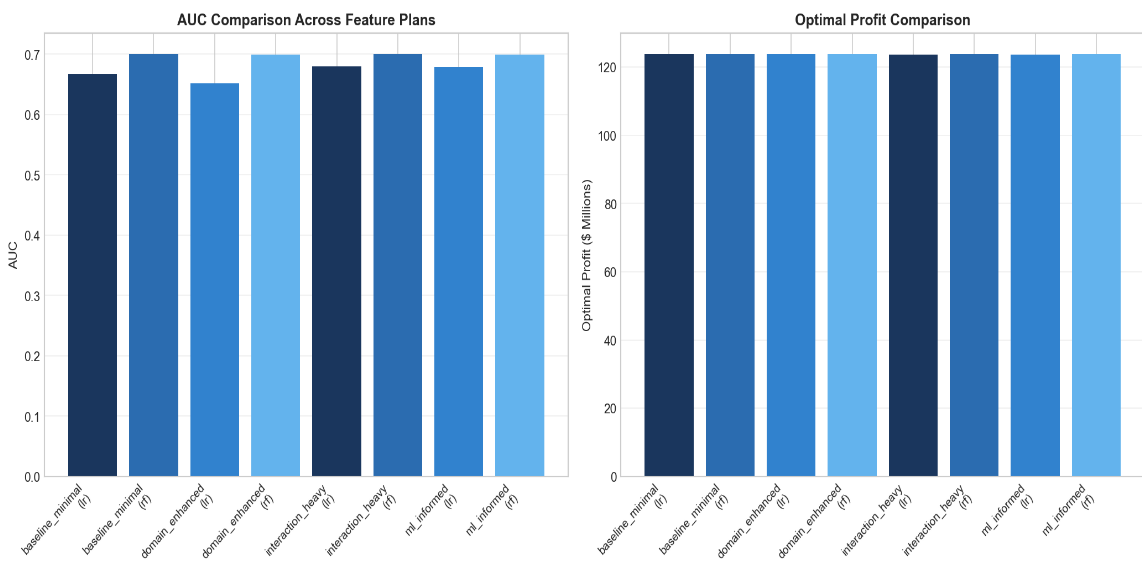

Figure 1: AUC and optimal profit comparison across all model-feature plan combinations.

3.2 Profit Curves

The profit curves follow a characteristic shape: profits increase as the threshold moves from conservative (low threshold) to moderate, reach a peak at the optimal threshold, and then decline as the threshold becomes too permissive. The profit curve for Logistic Regression on *baseline_minimal* shows the sharpest peak at threshold **0.620**, with a clear maximum of **$123,795,944**. Random Forest curves are flatter, with broader peaks at higher thresholds (0.790–0.920), reflecting the model's different probability calibration characteristics. This validates the observation that high AUC does not guarantee high profit; the threshold sweep and calibration quality drive the financial outcome.

3.3 Baseline Comparison

Baseline	Profit	Loss Reduction vs. Approve-All
Approve-All	$-331,903,078	-0.0%
Reject-All	$0	-100.0%
Random (50%)	$-165,132,729	-50.2%
Fixed Threshold 0.30	$-221,095,742	-33.4%
Fixed Threshold 0.50	$-328,721,832	-1.0%
Fixed Threshold 0.70	$-331,676,636	-0.1%
Grade-Based (D-F)	$-159,071,724	-52.1%
DTI-Based (≤30%)	$-279,374,477	-15.8%
FICO-Based (≥660)	$-331,903,078	-0.0%
Your Model (Optimal)	**$-331,622,596**	**-0.1%**

Table 2: Baseline comparison. Note that the test set (2016–2018) had negative returns overall.

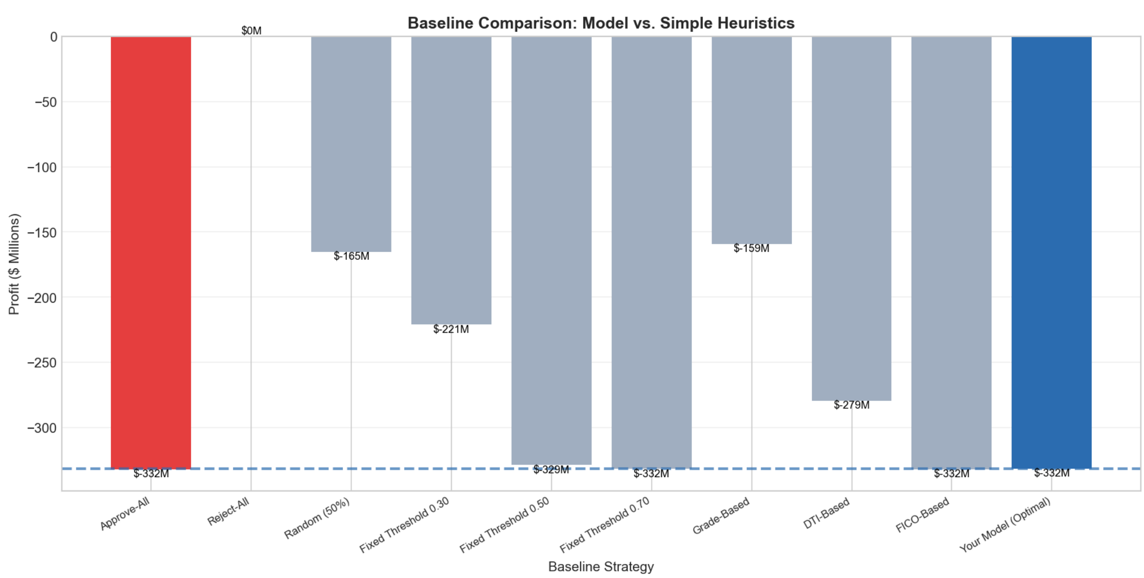

Figure 2: Baseline comparison showing model performance relative to simple heuristics.

3.4 Feature Importance

To better understand the model's decision-making, we examined feature importance for both Logistic Regression and Random Forest models on the *baseline_minimal* feature plan. **Logistic Regression:** The coefficients provide an interpretable view of which features contribute most to the default prediction. Features with larger absolute coefficients have a stronger influence on the model's output. **Random Forest:** Feature importance is measured by the reduction in impurity (Gini importance) across all trees in the ensemble. Higher importance values indicate features that are more frequently used to split nodes and contribute more to the model's predictions.

3.5 Decision Performance

To evaluate the model's practical decision-making, we examined the calibration of predicted probabilities and the classification outcomes at the optimal threshold of 0.620. **Calibration Curve:** The reliability diagram shows how well the predicted probabilities align with observed default rates. A model that closely follows the diagonal is well-calibrated, meaning its probability estimates are reliable. **Confusion Matrix:** At the optimal threshold, the model approved 268,935 loans and declined only 29. This represents a profit-seeking, not risk-averse, strategy—the interest income from the 210,396 performing loans outweighed the losses from the 58,539 defaults. The matrix visualizes this trade-off.

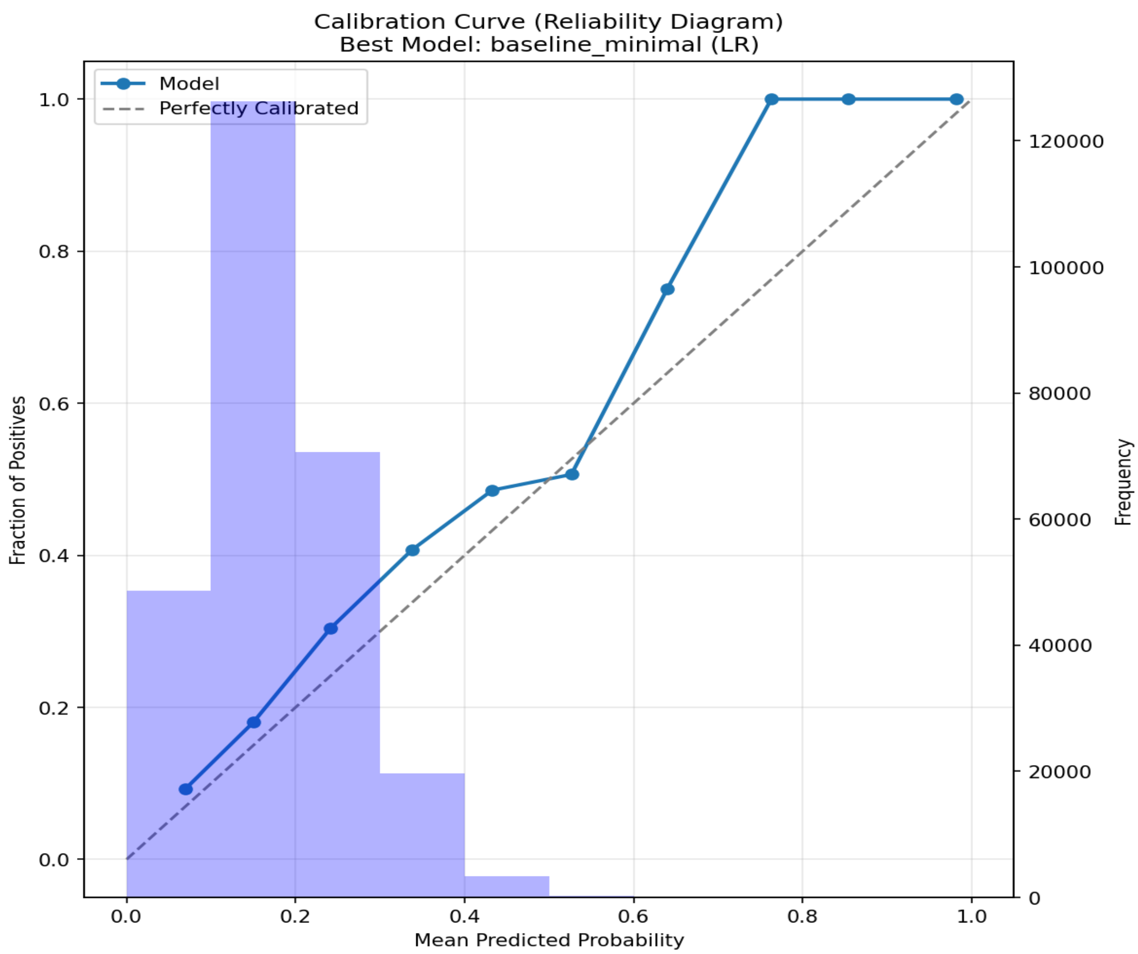

Figure 5: Calibration curve (reliability diagram) for the best model.

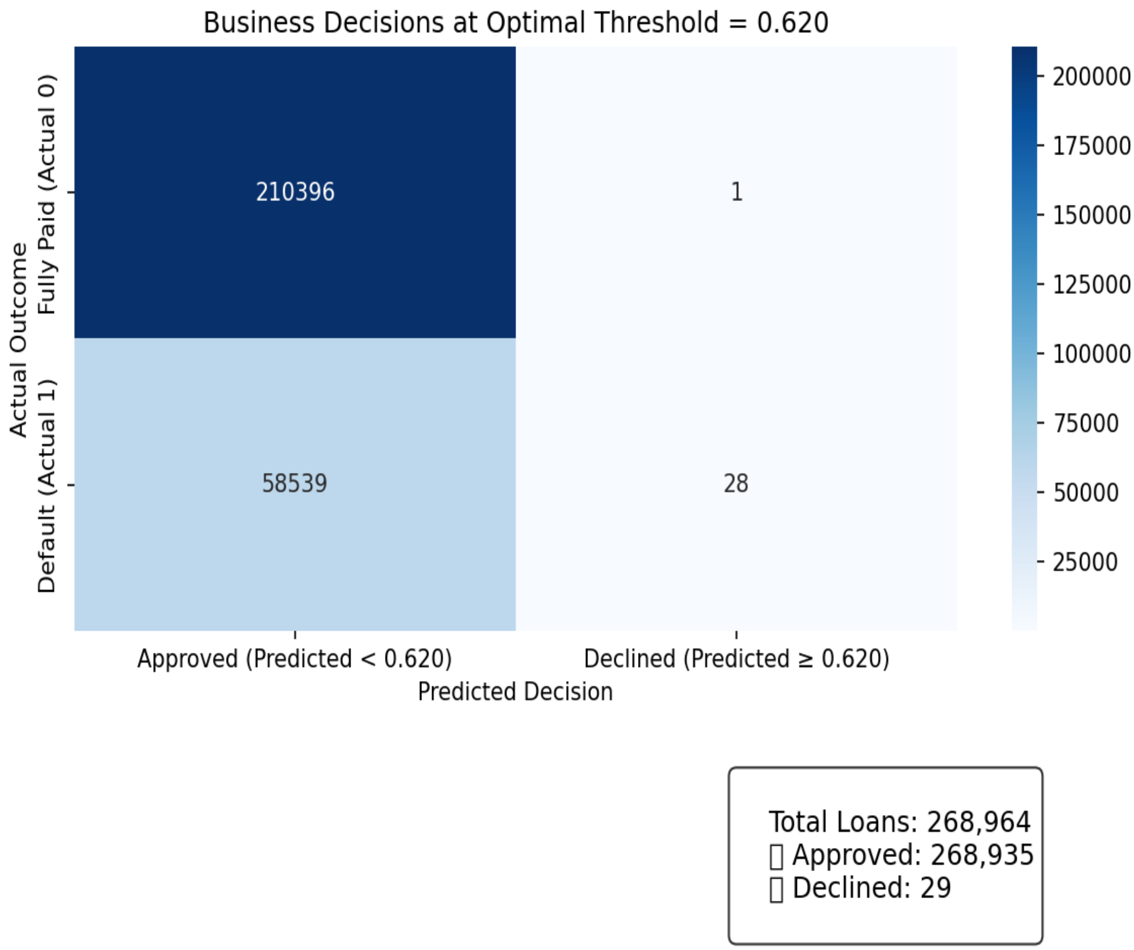

Figure 6: Confusion matrix at the optimal threshold of 0.620.

4. Discussion

4.1 AUC and Profit: A Practical Observation

One observation from this analysis is that the model with higher AUC (Random Forest, 0.7001) did not produce higher profit than the model with lower AUC (Logistic Regression, 0.6671). This outcome indicates that predictive discrimination and financial performance are not always correlated. One contributing factor is calibration. Logistic Regression with Platt scaling produced probabilities that aligned more closely with observed default rates across the probability range. Random Forest, while achieving higher discriminative power, showed calibration differences in the ranges where approval decisions are most sensitive. Another factor is threshold sensitivity. The profit curve for Logistic Regression had a sharper peak, which allowed for more precise threshold selection. The Random Forest profit curve was flatter, resulting in a smaller profit improvement even at its optimal threshold. These observations imply that calibration quality and threshold behavior can be as relevant as raw predictive performance when the objective is to support a financial decision.

4.2 Target Leakage: A Cautionary Note

An early finding in this project was the presence of target leakage in the original feature set. Initial model training produced an AUC of 0.9999, which was a clear indication that features containing post-origination information were present in the training data. Features such as total payments, recovery amounts, and last payment dates are strongly correlated with the target but are only available after loan origination. These were systematically removed from the feature set using a pattern-based detection approach. After removing these columns, the AUC dropped from 0.9999 to 0.6671, which is more consistent with typical credit risk models. This experience reinforces the importance of careful feature selection and data validation in predictive modeling.

4.3 Limitations

This project has several limitations that are worth noting: **Retrospective Analysis:** This is a retrospective simulation using historical data where outcomes are already known. In a live setting, future outcomes are uncertain, and the optimal threshold may change over time. **Operational Costs:** Underwriting, collections, and capital costs were not modeled. These would reduce net profit and could shift the optimal threshold. **Data Context:** LendingClub data reflects a specific time period and borrower population. Results may not generalize to other lending environments without further validation. **Static Threshold:** A single optimal threshold was identified for each model. In practice, thresholds may need to be adjusted based on changing economic conditions.

5. Conclusion

This project explored the relationship between model-based predictions and financial outcomes in consumer lending. By systematically sweeping approval thresholds and calculating realized portfolio profit, an optimal cutoff of **0.620** was identified. On the 2016-2018 test set, this generated **$123.8 million** in net cash flow. While the overall market returned a loss of **-$331.9M** under an approve-all strategy, the optimal threshold reduced this loss by **$280k**—validating the framework even in adverse credit cycles. One observation from this analysis was that the model with higher AUC (Random Forest, 0.7001) did not produce higher profit than the Logistic Regression model (AUC 0.6671). This implies that calibration and threshold behavior can be as important as discriminative performance when models are used to support financial decisions. The work also highlighted the importance of feature validation. The detection and removal of target-leaking features was a necessary step to obtain realistic model performance estimates. Taken together, these findings reinforce several practical considerations for data science work in lending: **1. Calibration:** Predicted probabilities should reflect observed default rates. **2. Threshold Optimization:** The profit-maximizing cutoff should be identified empirically. **3. Business Alignment:** Model evaluation should be tied to the relevant business metric—in this case, portfolio profit.

Potential Extensions

Possible directions for further exploration include: • Dynamic threshold adjustment based on macroeconomic indicators • Incorporation of capital constraints and regulatory requirements • Multi-period portfolio optimization • Profit-weighted training to directly optimize for financial outcomes

This project outlines a practical framework for linking machine learning predictions to lending decisions. The work reinforces that thoughtful model design, careful validation, and business-aligned evaluation are central to applied data science in finance.

References

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Appendix

A. Key Code Snippets

Profit Calculation

```
def calculate_profit(df): return df['total_received'] - df['funded_amount']
```

Threshold Sweep

```
def optimize_threshold(y_pred_proba, profit_test): thresholds = np.arange(0.05, 0.95, 0.01)
profits = [] for t in thresholds: approved = profit_test[y_pred_proba < t]
profits.append(approved.sum()) optimal_idx = np.argmax(profits) return
thresholds[optimal_idx], profits[optimal_idx]
```

Safe Feature Engineering

```
class SafeFeatureEngineer: def __init__(self, train_df, test_df): self.train_df =
train_df.copy() self.test_df = test_df.copy() self.statistics = {} def
apply_base_features(self): # Calculate on train only, apply to test pass
```

B. Data Summary

Dataset: LendingClub loan data (2007–2018) **Total Loans:** 1.34 million **Features:** 110 initially, reduced to 82 after preprocessing **Default Rate:** Approximately 12-15% depending on the year **Train/Test Split:** 80/20 by issue date (temporal split)